

pharmaniaga®

Cetirizine

 **Tablet**
 **Oral Solution**

DESCRIPTION

Pharmaniaga Cetirizine tablet 10 mg

White, oblong, film-coated tablet, plain on one side and scored on the other side

Pharmaniaga Cetirizine oral solution 0.1% w/v

Clear, colourless solution

COMPOSITION

Pharmaniaga Cetirizine tablet 10 mg

Each tablet contains: Cetirizine hydrochloride 10 mg

Pharmaniaga Cetirizine oral solution 0.1% w/v

Each mL contains: Cetirizine hydrochloride 1.0 mg

ACTIONS

Cetirizine is a potent H₁-selective antihistamine. It belongs to the piperazine class of antihistamines and is the major carboxylated metabolite of hydroxyzine. However, it is less lipophilic than hydroxyzine and thus less likely to distribute into the central nervous system and produce sedation. Cetirizine generally has greater selectivity for H₁-receptor, less anticholinergic side effects and less CNS penetration.

Cetirizine has shown that it will inhibit certain effects produced by exogenous histamine and this activity appears rapidly. Cetirizine also appears to inhibit the effects produced by endogenous histamine released in vivo. Cetirizine may also have a modulatory effect on inflammatory cells by inhibiting eosinophil migration or eosinophilotactic mediators. Cetirizine has demonstrated a significant reduction in skin eosinophil recruitment.

Additionally, cetirizine may also have inhibitory effects on other infiltrating inflammatory cells, such as neutrophils. The second generation antihistamines have been used in clinical trials for the adjunctive treatment of pollen-induced asthma. They are more selective for the histamine receptor and may inhibit other inflammatory processes that may be involved in asthma. Cetirizine has been shown to inhibit eosinophilic migration in wheal and flare skin challenge tests, thus offering a potential application in the treatment of asthma.

Cetirizine is rapidly absorbed with a time (T_{max}) of approximately 1 hour to maximum concentration following oral administration of tablets or syrup in adults. When healthy volunteers administered with multiple doses of cetirizine (10 mg tablets once daily for 10 days), a mean peak plasma concentration (C_{max}) of 311 ng / mL can be observed. No accumulation would be observed. Cetirizine pharmacokinetics are linear for oral doses ranging from 5 to 60 mg. Food has no effect on the extent of cetirizine exposure (AUC) but T_{max} is delayed by 1.7 hours and C_{max} is decreased by 23% in the presence of food. In one pharmacokinetic study, the oral absorption of an aqueous solution of cetirizine during fasting was estimated to be at least 70%.

Cetirizine is strongly bound to plasma proteins. It has been shown through animal studies that cetirizine has minimal penetration into the CNS and does not significantly interact with cerebral H₁ receptors. Cetirizine is not extensively metabolised by the liver. One minor metabolite has been identified in both the plasma and faeces but not in the urine. Approximately 60% of a 10 mg oral dose is excreted unchanged in the urine within 24 hours, with an additional 10% appearing in the urine over the next 4 days. The mean elimination half-life was found to be 8.3 hours from a multiple pharmacokinetic studies.

INDICATIONS

It is used for the symptomatic relief of allergic condition including rhinitis and chronic urticaria.

CONTRAINDICATIONS

Cetirizine is contraindicated in those patients with a known hypersensitivity to it or any of its ingredients or hydroxyzine. It is also contraindicated during lactation.

ADVERSE REACTIONS / SIDE EFFECTS

There have been reports of mild and transient side-effects such as headache, dizziness, drowsiness, agitation, dry mouth and gastrointestinal discomfort. The incidence of sedation with cetirizine was reported to be similar to that of placebo.

Occasionally symptoms of hypersensitivity have been reported.

Effects on ability to drive and operate machines

Cetirizine may have minor or moderate influence on the patient's ability to react. This should be considered when extra alertness is required e.g. when driving. Cetirizine may potentiate the effects of alcohol and CNS depressants. In sensitive patients, concurrent use with alcohol or other CNS depressants may cause additional reductions in alertness and impairment of performance.

PRECAUTIONS

Activities Requiring Mental Alertness: In clinical trials the

occurrence of somnolence has been reported in some patients taking cetirizine: due caution should therefore be exercised when driving a car or operating potentially dangerous machinery.

Concurrent use of cetirizine with alcohol or other CNS depressants should be avoided because additional reductions in alertness and additional impairment of CNS performance may occur.

The use of cetirizine in children below 6 years of age with impaired renal function or hepatic function should be avoided.

Safety and efficacy have not been established for children up to 2 years of age.

Pregnancy and lactation

As a precaution, cetirizine should not be administered to pregnant women during the first three months of pregnancy. Cetirizine is contraindicated in lactating women since it is excreted in breast milk.

Drug Interactions

Co-administration of theophylline and cetirizine may cause decreased cetirizine clearance resulting in elevated cetirizine serum concentrations and possibly cetirizine toxicity. Reductions in cetirizine doses may be required. Cetirizine should be used with caution if sedatives are also being taken.

DOSAGE AND ADMINISTRATION

Cetirizine oral solution 0.1% w/v

In children 6 years old or above: 10 mg daily, either 5 mg twice daily or 10 mg once daily.

(10 mg = 10 mL solution; 5 mg = 5 mL solution)

Cetirizine tablet 10 mg

In adults, the recommended dose is 10 mg once daily.

(10 mg = 1 tablet)

Note: In patients above 6 years of age and older with decreased renal function (creatinine clearance 11 to 31 mL/ min), patients on haemodialysis (creatinine clearance less than 7 mL/ min) and in hepatically impaired patients, a dose of 5 mg once daily is recommended.

OVERDOSAGE

Overdosage has been reported with cetirizine. In one adult patient who took 150 mg of cetirizine the patient was somnolent but did not display any other clinical signs or abnormal blood chemistry or haematology results. In a paediatric patient who took an overdose of cetirizine (approximately 180 mg), restlessness and irritability were observed initially; and was followed by drowsiness. Should overdose occur, treatment should be symptomatic or supportive, taking into account any concomitantly ingested medications. There is no known specific antidote to cetirizine. Cetirizine is not effectively removed by dialysis, and dialysis will be ineffective unless a dialysable agent has been concomitantly ingested.

Drowsiness can be a symptom of overdosage, occurring from administration of 50 mg of cetirizine as a single dose.

Since there is no specific antidote for overdose with antihistamines, treatment is symptomatic and supportive.

To decrease absorption - Induction of emesis (syrup of ipecac recommended); however, precaution against aspiration is necessary, especially in infants and children.

- Gastric lavage (isotonic or 0.45% sodium chloride solution) if patient is unable to vomit within 3 hours of ingestion.

To enhance elimination - Saline cathartics (milk of magnesia) are sometimes used.

ROUTE OF ADMINISTRATION

Oral.

STORAGE CONDITION

Store below 30°C.

Protect from light.

Keep container tightly closed.

SHELF LIFE

Product should not be used beyond the expiry date imprinted on the product packaging.

PRESENTATION

Cetirizine tablet 10 mg

In blisters of 500 tablets.

Cetirizine oral solution 0.1% w/v

In bottles of 60 mL.

Date of revision: 10th September 2020

PRODUCT REGISTRATION HOLDER / MANUFACTURER:

Pharmaniaga Manufacturing Berhad (198001006232)

No. 11A, Jalan P/1, Kawasan Perusahaan Bangi,
43650 Bandar Baru Bangi, Selangor Darul Ehsan, Malaysia.

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